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EXPERIMENT – 13

Study of Phishing Attacks Using Z-Phisher (Open-Source GitHub Tool) in Kali Linux – Educational Demonstration

Aim

To study phishing attack concepts using the Z-Phisher open-source framework in Kali Linux in order to understand social-engineering threats, user behavior exploitation, and preventive security measures.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Study Tool:** Z-Phisher (Open-Source GitHub Framework – Educational Use)
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

Phishing is a social-engineering attack where attackers trick users into revealing sensitive information such as usernames, passwords, or OTPs by impersonating trusted entities. Z-Phisher is an open-source educational framework that demonstrates how phishing pages are structured and how attackers exploit human trust rather than software vulnerabilities. Studying phishing attacks in a controlled Kali Linux environment helps learners understand attack workflows, risks, and the importance of security awareness and defense mechanisms.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Study the Z-Phisher Framework

1. Visit the Z-Phisher GitHub repository (read-only study).
2. Review documentation to understand:
   - Types of phishing templates
   - Social-engineering techniques used
   - Ethical usage warnings

Step 3: Understand Phishing Attack Workflow

1. Analyze how fake login pages mimic legitimate websites.
2. Observe how users are tricked into entering credentials.
3. Study the role of URL masking and trust exploitation.

Step 4: Controlled Demonstration (Simulation Only)

1. Perform a **local, consent-based simulation** using dummy accounts.
2. Observe how phishing pages appear to users.
3. Analyze user response behavior (no real credentials used).

Step 5: Identify Security Risks

- Credential theft
- Identity compromise
- Financial fraud
- Privacy violations

Step 6: Study Prevention Techniques

- User awareness and phishing education
- Browser security warnings
- Two-factor authentication (2FA)
- Email and URL filtering
- HTTPS and domain verification

Output

- Clear understanding of phishing techniques and attack flow.
- Identification of social-engineering risks.
- Screenshots and notes recorded for academic reference.

README GPL-3.0 license

Zphisher

VERSION 2.3.5 | LICENSE GPL-3.0 | STARS 16K | ISSUES 355 OPEN | FORKS 6K

Author htr-tech | Open Source Yes | Maintained? Yes | Written In Bash

A beginners friendly, Automated phishing tool with 30+ templates.

Disclaimer

*Any actions and or activities related to **Zphisher** is solely your responsibility. The misuse of this toolkit can result in **criminal charges** brought against the persons in question. **The contributors will not be held responsible** in the event any criminal charges be brought against any individuals misusing this toolkit to break the law.*

***This toolkit contains materials that can be potentially damaging or dangerous for social media**. Refer to the laws in your province/country before accessing, using,or in any other way utilizing this in a wrong way.*

***This Tool is made for educational purposes only**. Do not attempt to violate the law with anything contained here. **If this is your intention, then Get the hell out of here**!*

*It only demonstrates "how phishing works". **You shall not misuse the information to gain unauthorized access to someones social media**. However you may try out this at your own risk.*

Zphisher Release 2.3.5 Latest
on Jan 3, 2023

+ 9 releases

Contributors 15

Languages

HTML 90.1% CSS 6.6% PHP 2.7% Other 0.6%

:: Workflow ::

Terminal - sudo bash zphisher.sh

File Edit View Terminal Tabs Help

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      |_|                Version : 2.2

[-] Tool Created by htr-tech (tahmid.rayat)

[::] Select An Attack For Your Victim [::]

[01] Facebook      [11] Twitch       [21] DeviantArt
[02] Instagram     [12] Pinterest    [22] Badoo
[03] Google        [13] Snapchat     [23] Origin
[04] Microsoft     [14] Linkedin     [24] DropBox
[05] Netflix       [15] Ebay         [25] Yahoo
[06] Paypal        [16] Quora        [26] Wordpress
[07] Steam         [17] Protonmail   [27] Yandex
[08] Twitter       [18] Spotify      [28] StackoverFlow
[09] Playstation   [19] Reddit       [29] Vk
[10] Tiktok        [20] Adobe        [30] XBOX
[31] Mediafire     [32] Gitlab       [33] Github

[99] About         [00] Exit

[-] Select an option :
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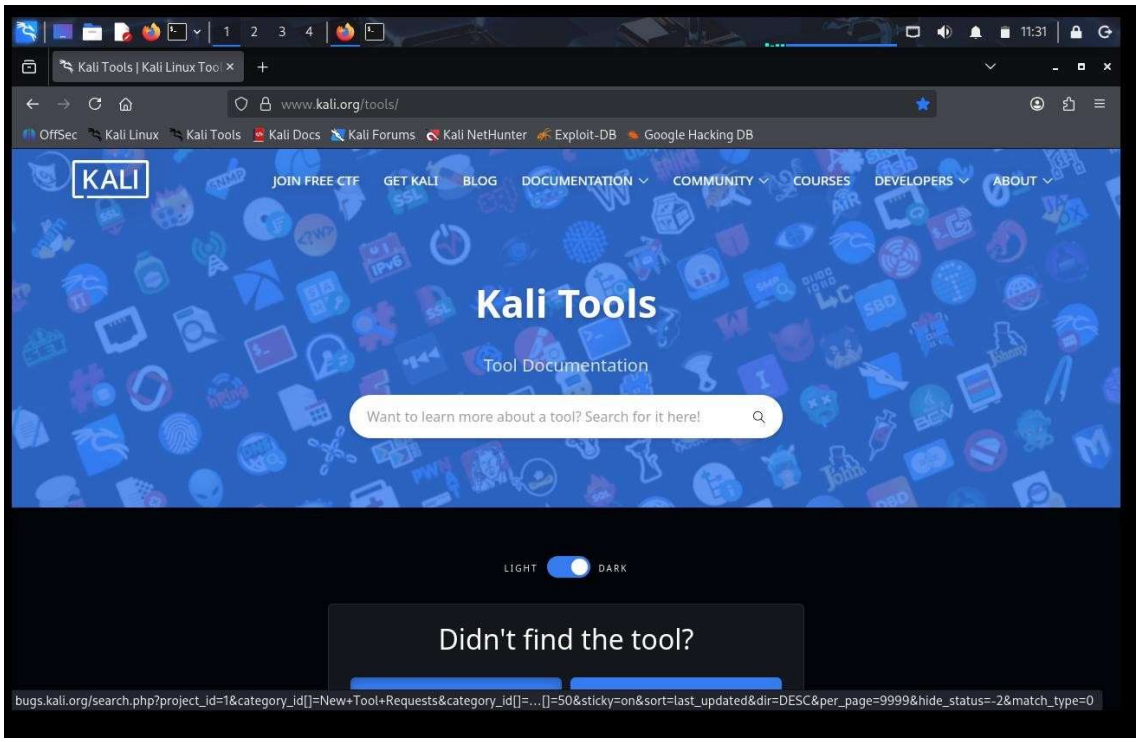

Result

The study of phishing attacks using the Z-Phisher framework was successfully completed in Kali Linux.
This experiment improved understanding of social-engineering threats and emphasized the importance of user awareness and defensive security practices.